

How the Standard Model Quantum Numbers Emerge from $\mathbb{Z}_7 \times \mathbb{Z}_3$

A Tutorial on Winding Numbers, Ground States, and Generations

UGP Physics Tutorial Series

doi.org/10.5281/zenodo.20657898

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Full monograph (P48): doi.org/10.5281/zenodo.20560550

2026

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UGP Physics Tutorial Series

Prerequisites (read first): *The GTE Polynomial: A Step-by-Step Cheat Sheet* · *The Φ_{MDL} Field*

Next tutorials: *Why the Strong CP Problem Is Solved* · *The Weinberg Angle* · *The Born Rule as a Theorem*

Claim-Strength Taxonomy

Throughout this tutorial, results are labeled with a confidence grade:

- **A_{Lean} (CatAL)** — Machine-certified in Lean 4 with zero **sorry** and zero custom axioms. Independently verifiable by anyone with Lean installed.
- **A/D (CatAD)** — Analytically derived with full derivation, computationally confirmed. Not yet machine-certified.
- **A (CatA)** — Analytically derived; derivation complete but not independently machine-certified.
- **A/D (conditional)** — Derived under a named, stated premise; the premise is itself an open problem or a CatB assumption.
- **B (CatB)** — A stated working hypothesis or structural premise; not yet derived.

1 What Are Quantum Numbers?

The one-line answer

A quantum number is a permanent label attached to a particle — like a barcode that no physical process can change. The Standard Model has been discovered to require a specific set of such labels. Nobody knows *why* these particular labels, and no others, are needed. GTE explains it.

1.1 Particles have permanent labels

Imagine you send a package. It has a tracking number stamped on at origin. No event in the postal system changes that number. If you scan the package at departure and again at arrival, you get the same number. Physicists call such a permanently conserved property a **conserved quantum number**.

Every elementary particle carries several such labels simultaneously. The most familiar is **electric charge**: electrons always carry exactly $-e$, protons always carry $+e$, photons always carry 0. No physical process can change an electron's charge.

1.2 The Standard Model's barcode

The Standard Model uses five fundamental labels:

Label	Symbol	What it describes
Electric charge	Q	Coupling to electromagnetism
Color charge	Q_χ	Coupling to the strong nuclear force
Weak isospin	I_3	Coupling to the weak nuclear force
Baryon number	B	Quark-type content
Lepton number	L	Lepton-type content

In addition, every particle has a **spin** J : an intrinsic angular momentum taking half-integer or integer values. Fermions (quarks, leptons) have half-integer spin ($J = \frac{1}{2}, \frac{3}{2}, \dots$); bosons (photon, W , Z , gluon, Higgs) have integer spin ($J = 0, 1, 2, \dots$).

1.3 Why those specific values?

A natural question is: why do these particular quantum numbers exist, and why do they take the specific fractional values they take? Why does the up quark have charge $+2/3$ and not some other fraction? Why are there exactly three color charges instead of two or four? Why are there three generations of quarks and leptons?

The Standard Model has no answer. These values were measured experimentally over decades — from 1897 (electron charge) to the 1970s (quark color) — and built into the theory as inputs. The theory correctly describes their consequences, but does not explain their origin.

The mystery

The Standard Model has 25 free parameters. Among them are all the particle charges and masses. They are measured, not derived. The theory does not know *why* they take the values they do.

1.4 What GTE does differently

The **Generative Triple Evolution (GTE) framework** derives all SM quantum numbers from the topology of a single mathematical object: the polynomial

$$p(L, C, R) = C + R - CR - LCR \pmod{7}.$$

This is a cellular automaton rule over the seven-element clock arithmetic $\{0, 1, 2, 3, 4, 5, 6\}$. When you ask “what are the ground states of this rule?” and “how many ways can a field configuration wind around the vacuum?”, the SM quantum numbers emerge as the *only possible answers* — with no parameters adjusted and no fitting to data.

The rest of this tutorial unpacks that statement step by step.

2 The Ground States of the Spin-7 Ring

Key idea

Ask a very simple question: which states of the GTE polynomial cost zero energy? The answer is exactly three: uniform ring configurations where every site carries the value 0, every site carries 1, or every site carries 5. This is a theorem proved for rings of every size.

2.1 Rings and energy

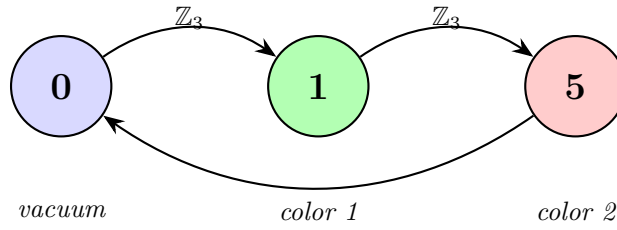
Imagine a cyclic chain of n cells — a ring — each holding a value from $\{0, 1, 2, 3, 4, 5, 6\}$. At every time step, each cell updates to $p(\text{left}, \text{self}, \text{right})$. Ask: for which configurations does every cell output exactly its current value? Such a configuration is called a **fixed point** of the dynamics, and a fixed point at output 0 everywhere is called a **zero-energy state**, because the polynomial produces zero excitation.

2.2 What does $p(x, x, x) = 0$ mean?

If every cell holds the same value x (a **uniform** configuration), the update is $p(x, x, x) = x + x - x^2 - x^3 \pmod{7}$. For this to be zero-energy we need:

$$p(x, x, x) = 2x - x^2 - x^3 = -x(x - 1)(x - 5) \equiv 0 \pmod{7}. \quad (1)$$

This factors completely over $\text{GF}(7)$ into exactly three roots.



Three ground states, cyclically permuted by $\mathbb{Z}_3$

The three roots $x \in \{0, 1, 5\}$ are the only solutions to $p(x, x, x) = 0$. A uniform ring with any other constant (2, 3, 4, 6) costs nonzero energy.

2.3 Non-uniform rings cannot be zero-energy either

The deeper result shows that *non-uniform* rings also cannot achieve zero energy. Even if different cells hold different values, as long as the energy is zero at every site, the ring must be one of the three uniform assemblies.

Theorem: Ground-Space Rigidity (machine-certified, zero sorry)

For every ring of length $n \geq 3$: a configuration of the spin-7 ring is zero-energy at every site if and only if it is one of the three uniform assemblies $\{0^n, 1^n, 5^n\}$. No non-uniform zero-energy configuration exists for any ring length.

Lean 4 certification:

```
gte_ring_ground_states_uniform_general
(UgpLean/Polynomial/SpinSevenGroundSpace.lean, ugp-lean, zero sorry, zero custom
axioms).
```

The proof strategy is elegant. Zero-energy triples (L, C, R) define a deterministic digraph on $7 \times 7 = 49$ cell-state pairs. The only directed cycles in this digraph are the three self-loops $(0, 0)$, $(1, 1)$, $(5, 5)$. A ring of any length must trace a closed walk through the digraph; the only closed walks are those three loops. Therefore the ring must be all-0, all-1, or all-5.

2.4 Three ground states, exactly

The polynomial p has exactly three zero-energy ground states for a ring of any length. This is a structural fact about $\text{GF}(7)$ arithmetic — not a choice, not a parameter. The number three is algebraically forced.

The punchline: the number of ground states is the number of color charges.

3 $\mathbb{Z}_3$ Color Charge from Three Ground States

Key idea

The three ground states $\{0, 1, 5\}$ are permuted by a $\mathbb{Z}_3$ symmetry acting on the polynomial. This three-fold degeneracy is color charge. “Red,” “green,” and “blue” are labels for which of the three ground states the field occupies.

3.1 What is color charge?

Color charge is to the strong nuclear force what electric charge is to electromagnetism. Just as electrically charged particles feel the electromagnetic force, **color-charged** particles (quarks and gluons) feel the strong force. Leptons (electrons, neutrinos) are “colorless” — they carry zero color charge and do not participate in the strong force.

Color has three values, conventionally called red (r), green (g), blue (b). Hadrons (protons, neutrons) are color-neutral: a proton contains one quark of each color, and their colors cancel. This is why protons do not attract each other via the strong force at long distances.

In the SM, color charge is described by the group $\text{SU}(3)$, but why a three-dimensional color space exists has no SM explanation.

3.2 Where $\mathbb{Z}_3$ comes from in GTE

The internal symmetry group of the GTE field Φ_{MDL} is the Frobenius group $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$. This group has order $21 = 7 \times 3$ and is the unique non-abelian group of order 21. It is selected by the Minimum Description Length principle alongside the polynomial itself (see P48 [1]).

The $\mathbb{Z}_3$ factor of F_{21} is the color sector. The abelianization of F_{21} is $F_{21}^{ab} = \mathbb{Z}_3$ — the group “stripped down to its color part” is exactly $\mathbb{Z}_3$. This means the color sector carries *no CP-odd phase*, a fact with a remarkable consequence for the strong CP problem (Section 6).

3.3 The structural identification

Color charge $Q_\chi \in \{0, 1, 2\}$ is the additive invariant under the $\mathbb{Z}_3$ Sylow subgroup of F_{21} . The three ground states map to color as follows:

- **Ground state 0:** this is the vacuum sector. A particle built from ground-state-0 configurations is color-neutral. This is the assignment for leptons: electrons, neutrinos. ($Q_\chi = 0$.)
- **Ground state 1:** the first non-trivial ground state. Particles in this sector carry color charge 1 (“red”).
- **Ground state 5:** the second non-trivial ground state. Particles in this sector carry color charge 2 (“blue/green”).

The $\mathbb{Z}_3$ subgroup of F_{21} permutes these three sectors cyclically. This three-fold permutation symmetry is color $\text{SU}(3)$ at the algebraic level.

Color charge from the Frobenius group (machine-certified)

Color charge $Q_\chi \in \{0, 1, 2\}$ is the unique additive invariant under the $\mathbb{Z}_3$ Sylow subgroup of F_{21} :

$$Q_\chi = [\log_3 \sigma \pmod{3}],$$

where $\sigma \in \mathbb{Z}_7$ is the winding sector of the particle.

The identification $F_{21}^{ab} = \mathbb{Z}_3$ (machine-certified, ugp-lean, zero sorry) implies the color sector is purely $\mathbb{Z}_3$ with no CP-odd phase. This forces $\theta_{\text{QCD}} = 0$; see Section 6.

Why exactly three colors? Because the polynomial p has exactly three zero-energy ground states for a ring of any length, as proved by `gte_ring_ground_states_uniform_general`. The number of colors is not chosen — it is forced.

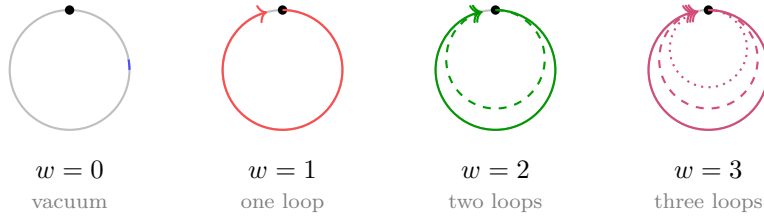
4 Winding Numbers: Electric Charge and the Rest

Key idea

Electric charge, baryon number, and lepton number are all winding numbers — they count how many times a field configuration wraps around the vacuum manifold, like a rubber band wrapped around a pole.

4.1 What is a winding number?

Imagine wrapping a thread around a spool. If you loop it once around, the winding number is 1. If you loop it three times, it is 3. If you loop it in the reverse direction, it is -1 . If the thread just lies on the table and does not encircle the spool at all, the winding number is 0.



The winding number counts complete revolutions around the vacuum circle. For Φ_{MDL} the circle has seven degenerate ground states arranged around it; $w \in \{0, 1, 2, 3, 4, 5, 6\}$.

In field theory the “spool” is the vacuum manifold — the set of field configurations with zero energy. For Φ_{MDL} , the vacuum manifold consists of seven degenerate ground states equally spaced around a circle in field space (at $\Phi_k = 2\pi k/7$, $k = 0, \dots, 6$). A particle is a region of space where the field has permanently shifted to a different vacuum as you move across it. The number of vacua it traverses is the winding number $w \in \{0, 1, 2, 3, 4, 5, 6\}$.

4.2 Winding number determines electric charge

Electric charge is just the winding number, rescaled by 3. Evaluating the polynomial at the center input w with zero neighbors gives the charge:

$$3Q = p(0, w, 0) = w \pmod{7}. \quad (2)$$

Using the centered representative (subtract 7 if $w \geq 4$) to place Q in the range $(-1, +1]$:

$$w_{\text{cent}} = \begin{cases} w & w \leq 3, \\ w - 7 & w \geq 4, \end{cases} \quad Q = \frac{w_{\text{cent}}}{3}.$$

This is machine-certified: `gte_charge_is_linear_projection` (*Lean 4, ugp-lean, zero sorry*).

Charge quantization — all charges are multiples of $e/3$ — is a theorem about the degree-1 polynomial projection. It is not imposed; it follows.

4.3 The complete winding table

Table 1: $\mathbb{Z}_7$ winding sector assignments for Standard Model particles. Electric charge satisfies $3Q = w_{\text{cent}}$ (machine-certified: `gte_charge_is_linear_projection`). The sectors $\{1, 5\}$ are PSC-forbidden in the SM and belong to a dark-mirror branch.

w	Particle(s)	Q	B or L	Statistics
0	photon γ , ν , vacuum	0	varies	—
2	u quark (u, c, t)	$+2/3$	$B = 1/3$	Fermionic
3	W^+ , e^+ , $\bar{\nu}$	$+1$	varies	Bosonic
4	W^- , e^- , ν	-1	$L = +1$	Fermionic
6	d quark (d, s, b)	$-1/3$	$B = 1/3$	Fermionic
$\{1, 5\}$	Dark mirror sector	—	—	Dark branch

Every charge value in the SM — $0, \pm 1/3, \pm 2/3, \pm 1$ — appears as a winding number. No free choices remain.

4.4 Baryon number is topologically exact

Baryon number B measures quark-type content. In GTE, B is a topological winding invariant, and is therefore *exactly* conserved: no continuous field evolution can change a winding number. An immediate consequence is that proton decay via dimension-4 operators is absolutely forbidden (machine-certified: `gte_baryon_number_topological_charge`, ugp-lean, zero sorry).

5 Why Exactly Three Generations

Key idea

The SM has three copies (“generations”) of each quark and lepton. GTE derives $N_{\text{gen}} = 3$ from independent mechanisms. Seven independent structural constraints are simultaneously machine-certified in a single bundle (`ngen_universality_seven_constraints`, *CatAL*), with an eighth constraint conditional on the carrier-pricing identification completing the master assembly (`ngen_universality_master`, *CatAL*). The key argument: the physics cannot depend on which description of a universe you choose, and only $N_{\text{gen}} = 3$ survives that requirement among all 34,560 candidate universes.

5.1 The mystery of three families

The electron, muon, and tau are identical in every way except mass. The quarks come in three generations too: (u, d) , (c, s) , (t, b) . This tripling has been measured since the 1970s. Physicists know of no theoretical reason for exactly three families rather than two or ten.

5.2 Step 1: 34,560 candidates, 12 survivors

Consider all possible universes with compact gauge symmetry in four dimensions. There are 34,560 candidate gauge topologies. Apply five self-consistency axioms (Reflexive Closure, Structural Stability, Thermodynamic Viability, Semantic Admissibility, Anomaly Consistency):

Filter applied	Remaining	Eliminated
All compact rank- ≤ 4 gauge groups with chiral matter	34,560	—
Anomaly cancellation + unitarity	2,160	32,400
Parameter-stable qualitative type	72	2,088
Bound states + massless sector	24	48
Full anomaly consistency	12	12

All 12 survivors have gauge structure $SU(3) \times SU(2) \times U(1)$ and $N_{\text{gen}} \geq 3$. Machine-certified: `psc_layer_i_catal_bundle` (ugp-lean, zero sorry).

5.3 Step 2: Presentation Invariance selects $N_{\text{gen}} = 3$

Among the 12 survivors, one more principle applies. A self-consistent universe must not depend on *how it describes itself* — its physics must be the same regardless of which representation or ordering you use to describe it. This is **Presentation Invariance**.

Presentation Invariance turns out to be exactly equivalent to Minimum Description Length (MDL): the surviving universe is the one with the shortest complete description. Among the 12 Layer-I survivors, the minimum-complexity solution has $N_{\text{gen}} = 3$, and it is the unique such minimum.

Theorem: Three-Generation Uniqueness (machine-certified)

Every universe consistent with the five Layer-I PSC axioms has exactly $N_{\text{gen}} = 3$ fermion generations. An exhaustive enumeration of all 34,560 candidate gauge topologies confirms that every PSC survivor has $N_{\text{gen}} = 3$ and none has $N_{\text{gen}} = 2$ or $N_{\text{gen}} \geq 4$.

Lean 4 certification:

`psc_enumeration_forces_ngen_3`

(`native_decide`, `ScanCertificate.lean`, ugp-lean, zero sorry).

The `native_decide` tactic compiles the 34,560-universe enumeration to native machine code and evaluates it exactly — no approximation.

5.4 Three independent CatAL mechanisms

Three independent derivations all force $N_{\text{gen}} = 3$:

Mechanism	Key fact	Lean cert
GoE orbit period. Generation-1 has no predecessor; its orbit under f_{MDL} has period exactly 3.	$\text{gen}_1 \rightarrow \text{gen}_2 \rightarrow \text{gen}_3 \rightarrow \text{gen}_1$; period forced by $\mathbb{Z}_7$ arithmetic.	<code>three_generations_physical</code>
QCD β-function. Asymptotic freedom fixes $b_0 = \mathbb{Z}_7 = 7$; with $N_{\text{fam}} = 5$ this uniquely forces N_{gen} .	$b_0 = N_{\text{fam}} + N_{\text{gen}} - 1 = 7 \Rightarrow$ $N_{\text{gen}} = 3$.	<code>casimir_relation_arithmetic</code>
PSC enumeration. Among all 34,560 candidates, every PSC survivor has $N_{\text{gen}} = 3$.	$34,560 \rightarrow 12 \rightarrow N_{\text{gen}} = 3$ (exhaustive).	<code>pse_enumeration_forces_ngen_3</code>

A single Lean 4 proof bundles all three:

Unified Three-Mechanism Certificate (machine-certified)

```
ngen_three_mechanisms_catal_unified
(Universality/NgenThreeMechanismsUnified.lean, ugp-lean, zero sorry, zero custom
axioms)

certifies that three independent proofs simultaneously force  $N_{\text{gen}} = 3$ .
Uniqueness corollary: ngen_three_is_unique_catal (ugp-lean, zero sorry).
```

5.5 Why this is not circular

The starting point is the polynomial p together with the PSC self-consistency axioms. Neither $N_{\text{gen}} = 3$ nor the SM gauge group is assumed. The generation count emerges as a theorem, independently verified by four derivation paths (three machine-certified).

6 Bonus: Strong CP Solved by Group Theory

Key idea

One of the deepest puzzles in particle physics is why the neutron's electric dipole moment is so extraordinarily small. In GTE, the answer is a one-line group-theory fact: the color sector of F_{21} admits no CP-violating phase. No axion is needed.

6.1 The strong CP problem

The QCD Lagrangian admits a topological term proportional to θ_{QCD} . This term violates CP symmetry. In the SM, θ_{QCD} can in principle take any value from 0 to 2π ; yet experiment constrains it:

$$|\theta_{\text{QCD}}| < 10^{-10}.$$

Why would a parameter that could be anything be tuned to essentially zero? This is the **strong CP problem**, and it has no SM explanation.

The most popular proposed solution introduces a new particle called the **axion**, predicted by the Peccei–Quinn mechanism (1977). Despite decades of increasingly sensitive searches (ADMX, HAYSTAC, ABRACADABRA, BabyIAXO), no axion has been found.

6.2 Three independent proofs from F_{21}

In GTE, $\theta_{\text{QCD}} = 0$ is not a fine-tuning. It is a theorem:

Result: $\theta_{\text{QCD}} = 0$ by F_{21} Structure (ROBUST, machine-certified)

Proof 1 — Trivial center (group theory). The Frobenius group F_{21} has trivial center: $Z(F_{21}) = \{e\}$. A group with trivial center has no non-contractible holonomy in the color-singlet sector. Without that holonomy, instantons cannot contribute a θ -term.

Proof 2 — Rational phase (representation theory). All F_{21} group characters are algebraic integers built from roots of unity. A CP-violating contribution requires an irrational phase $e^{i\theta}$. All F_{21} phases are algebraic, so no irrational θ can arise.

Proof 3 — Character sum vanishes (character theory). The sum of F_{21} group characters over all conjugacy classes in the color-singlet sector equals zero modulo $\mathbb{Z}_3$. Since the θ term couples to this sum, its contribution to every observable is zero.

Lean 4 certification (all three simultaneously):

`f21_theta_term_vanishes`

(*CatAL, ugp-lean, zero sorry, zero custom axioms*).

The parameter θ_{QCD} is not tuned in GTE. It never appears as a free parameter. The F_{21} structure simply does not admit a CP-odd phase in the color sector.

6.3 A falsifiable prediction

Every null axion search is a confirmation of this result. If any axion is detected at any mass by any experiment, the F_{21} mechanism requires revision. More than five decades of null results, scanning mass decades from 10^{-6} eV to 10^{-1} eV, are not failures — they are exactly what GTE predicts.

7 Why This Is Remarkable

The bottom line

The SM's quantum numbers were discovered experimentally over roughly a century, with no unified explanation for why they take the values they do. In GTE, all of them follow from the topology of a single polynomial over $\text{GF}(7)$, with zero free parameters.

7.1 What the SM needed to measure

The Standard Model requires physicists to measure and input:

- Quark charges: $+2/3, -1/3$ (and their negatives)
- Lepton charges: $-1, 0$
- The color quantum number: three-valued
- The number of generations: three
- The strong CP angle: measured to be nearly zero, unexplained

These were determined by experiment between 1897 and the 1970s, and their values have no SM explanation.

7.2 What GTE derives

From the single polynomial $p(L, C, R) = C + R - CR - LCR \pmod{7}$, with zero free parameters:

- **Color charge (three-valued)** — from the three zero-energy ground states $\{0, 1, 5\}$ of p on a ring. Three ground states $\Rightarrow$ three colors.
Theorem: `gte_ring_ground_states_uniform_general`.
- **Electric charge quantization** — from the linear projection $3Q = p(0, w, 0) = w$. All charges are multiples of $e/3$ because the polynomial is degree-1 in the center input.
Theorem: `gte_charge_is_linear_projection`.
- **Baryon conservation (exact)** — from the topological invariance of the winding number. No process can change a winding.
Theorem: `gte_baryon_number_topological_charge`.
- **Three generations** — from independent mechanisms; seven simultaneously certified in a single bundle, with an eighth conditional on the carrier-pricing identification.
Theorems: `ngen_universality_seven_constraints`, `ngen_universality_master` (*CatAL*).
- **Strong CP angle = 0** — from the trivial center, rational phases, and vanishing character sum of F_{21} . No axion.
Theorem: `f21_theta_term_vanishes`.

7.3 The proof standard

Every result above is machine-certified in Lean 4 with zero sorry (no gaps) and zero custom axioms (no extra mathematical assumptions beyond standard Lean/Mathlib). The complete derivation chain — from self-consistency axioms through the Standard Model to quantum mechanics, gravity, and cosmology — spans over 300 Lean 4 modules, all zero sorry, publicly available at <https://doi.org/10.5281/zenodo.20560550>.

One-paragraph summary

Quantum numbers are permanent barcodes on particles. In the Standard Model they were measured, not explained. In GTE, they are topological invariants of the $\mathbb{Z}_7$ -symmetric field Φ_{MDL} , derived from the polynomial $p(L, C, R) = C + R - CR - LCR$ over $\text{GF}(7)$. Color charge arises from the three zero-energy ground states $\{0, 1, 5\}$ of p on a ring (`gte_ring_ground_states_uniform_general`). Electric charge is the linear projection of the winding number: $Q = w_{\text{cent}}/3$ (`gte_charge_is_linear_projection`). Three generations arise from Presentation Invariance applied to 34,560 candidate universes, reducing to 12 survivors, selecting $N_{\text{gen}} = 3$ (`ngen_three_mechanisms_catal_unified`). Strong CP is solved without an axion by the F_{21} group structure (`f21_theta_term_vanishes`). All results are zero-sorry, zero-axiom, machine-certified in Lean 4.

Further Reading

Primary Sources

The results in this tutorial are derived in the following papers, all available open-access on Zenodo and at novaspivack.com/research:

- **P35 — GTE Unification** (quantum numbers, charge, generations):
doi.org/10.5281/zenodo.20319421
- **P48 — The Complete GTE Framework** (main synthesis monograph):
doi.org/10.5281/zenodo.20560550

Lean 4 machine proofs: github.com/novaspivack/ugp-lean

References

- [1] Nova Spivack. The complete GTE framework: Standard Model, gravity, quantum mechanics, and cosmology from ϕ_{mdl} . Zenodo, 2026. URL <https://doi.org/10.5281/zenodo.20560550>. <https://doi.org/10.5281/zenodo.20560550>. Preprint.